

PROJECT AND TEAM INFORMATION

Project Title

RouteWise: An Intelligent Energy-Aware Fuel and Charging Station Recommendation System

Student / Team Information

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PROPOSAL DESCRIPTION

The motivation behind this project is to help drivers make economical and informed decisions when refueling or charging during their trips. Drivers often find themselves with fuel or low battery levels and may not be sure if they can reach the next available station. Sometimes the closest station is not the affordable or it might have long waiting times.

RouteWise is designed to solve this problem by taking into account key factors at once—vehicle range, the current route the distance, to stations, fuel or charging prices waiting times and station availability. The system supports vehicles, petrol vehicles, diesel vehicles and hybrid vehicles. It provides alerts and smart station recommendations to help drivers avoid running out of fuel or battery. This helps users save both time and money while traveling.

State of the Art / Current solution

The current solution focuses on finding fuel or charging stations and offers basic navigation. Most existing systems let users find stations to where they are and give details like distance, price, availability or how long they might have to wait.

These systems often don't take into account the vehicle's current fuel or battery level. They may not consider how much range is left, the route the driver plans to take or the total cost and waiting time combined. They also tend to focus on electric vehicles or certain types of vehicles, which limits their usefulness, for other kinds of drivers.

Project Goals and Milestones

The main goal of the proposed RouteWise system is to create an easy-to-use solution for suggesting the best fuel and charging stations during a trip. The system will work with vehicles, petrol cars, diesel cars and hybrid vehicles. It will take into account the vehicle's fuel or battery level how much range is left, the distance to each station the price of fuel or charging any waiting time whether a station is available and the total estimated cost. The system will also alert users in advance if the vehicle might not make it to a station. This project aims to help drivers make choices save money and have better control, over their refueling or charging plans.

Project Milestones:

The project will be completed in steps.

- a) We will clearly define the problem set the objectives and finalize the features that the system must include.*
- b) Then we will design the system architecture plan the database structure and create the object-oriented programming classes.*
- c) After that we will build the core modules: the vehicle module, the station module, the distance calculation module the range-checking module and the recommendation engine.*
- d) Next we will add the warning system and the station comparison feature.*
- e) A user interface will then be developed to show details, station data, waiting times prices and suggested options.*
- f) Finally all parts will be. Tested using different vehicle types and travel routes.*
- g) We will evaluate performance, write documentation and prepare the final presentation.*

Project Approach

The proposed RouteWise system will use an step-by-step approach. First the user will input the location, the destination, the type of vehicle and the current fuel level or battery level. The system will then look for fuel stations. Charging stations that are along the route or close to it.

Next the system will calculate how far each station is from the position. It will also estimate whether the vehicle has fuel or battery to reach each station. Any station that cannot be reached safely will be removed from the list. If the vehicle's range is not enough to reach the closest station a warning will appear.

For the stations that're reachable the system will compare several factors: the distance to the station, the fuel or charging price the waiting time how many charging points are available and the total estimated cost. A weighted scoring method will be used to rank the stations. The system will rely on data structures, like priority queues, regular queues and hash maps to manage and sort this information quickly.

System Architecture

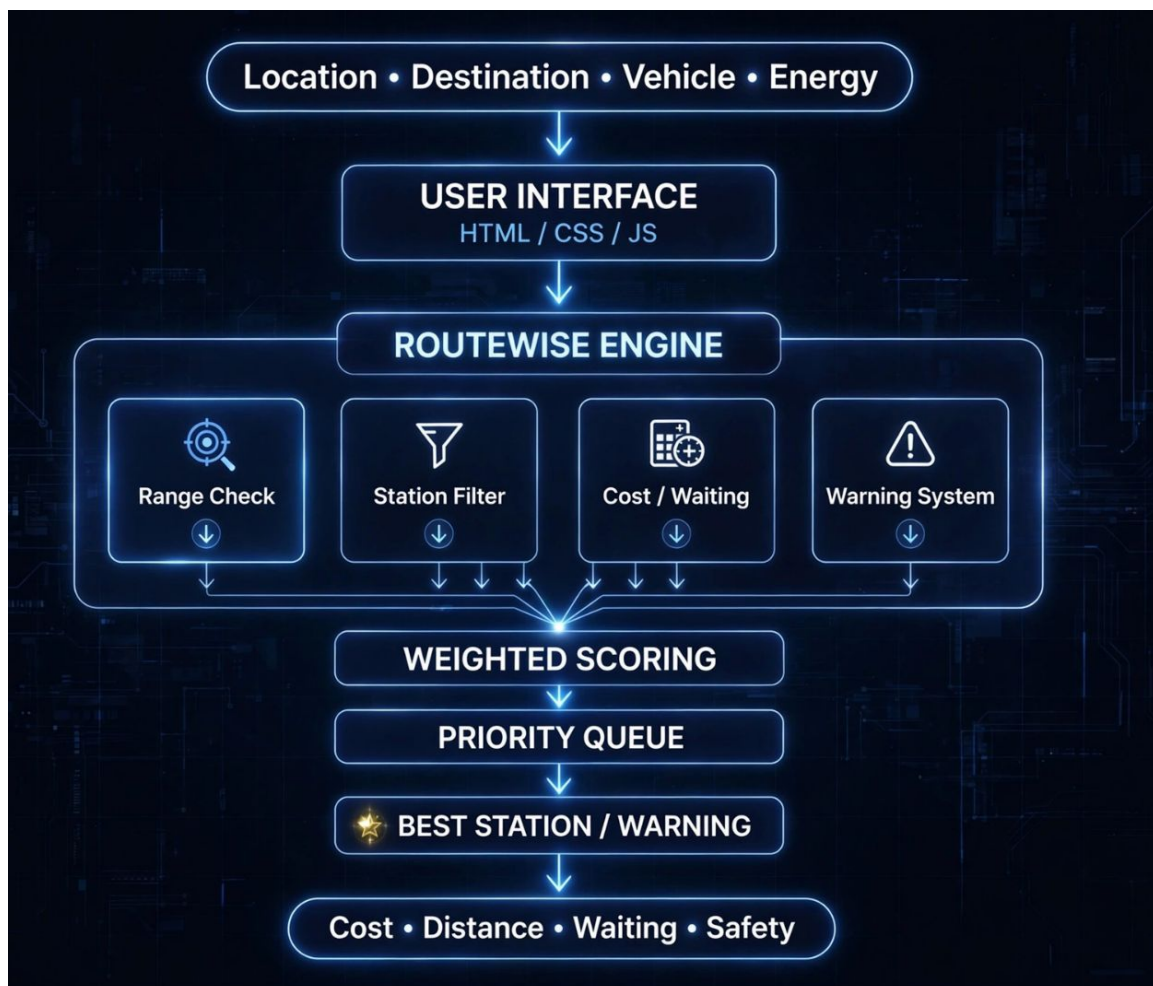


Fig- 1. System flowchart

Project Outcome

The expected outcome of the RouteWise project is an user-friendly Web Application that helps drivers select a suitable fuel or charging station during their journey. The system will support EV, petrol, diesel and hybrid vehicles. Consider their current fuel or battery level before making a recommendation.

The system is expected to:

- a) Calculate the vehicle's remaining driving range.*
- b) Identify fuel and charging stations along the journey.*
- c) Provide a warning when the vehicle may not reach a suitable station.*
- d) Compare stations based on distance, price, waiting time, availability and estimated cost.*
- e) Recommend the suitable and economical station.*
- f) Help users save time and money while reducing the risk of running out of fuel or battery.*
- g) Demonstrate the application of OOP, DSA, algorithms and database concepts, in a real-world problem.*

Assumptions

The RouteWise project is built on the following assumptions:

- a) The user gives a current location and a clear destination.*
- b) The user selects the type of vehicle: EV, petrol, diesel or hybrid.*
- c) The fuel or battery level entered is considered to be fairly accurate.*
- d) Information about stations including where they're how much they cost if they are available and how long the wait is is assumed to be present and mostly current.*
- e) The system looks at stations that're on or close to the planned trip route.*
- f) The energy use of the vehicle is expected to be roughly predictable based on the vehicle type and the energy it has.*
- g) The suggestions made are based on things, like if the driver can reach the station how far it is, how much it costs how long the wait. If there are spaces.*
- h) If real-time data is not there the system might use example or fake station data for testing purposes.*
- i) The system works under road and traffic conditions unless it has real-time traffic data.*
- j) The system gives suggestions and alerts. The driver makes the final choice.*

References

- a) Electric vehicle routing with heterogeneous charging stations. (2026). Computers & Operations Research, 189, 107374. The study considers charging price, charging technology, and waiting time in EV routing decisions.*
- b) Optimizing electric vehicle charging routes with deep reinforcement learning considering driver preferences. (2026). Sustainable Energy, Grids and Networks, 47, 102353.*
- c) Optimal charging facility location and capacity for electric vehicles considering route choice and charging time equilibrium. (2020). Computers & Operations Research, 113, 104776.*
- d) Dastpak, M., et al. (2024). Dynamic routing for the Electric Vehicle Shortest Path. This is especially relevant because it considers re-optimizing EV routes and charging decisions when system information changes, which supports your real-time recommendation idea.*
- e) Ferraz, R. S. F., et al. (2025). Optimized Planning for Electric Vehicle Charging Stations. This is a recent review of research on EV charging-station planning and optimization.*